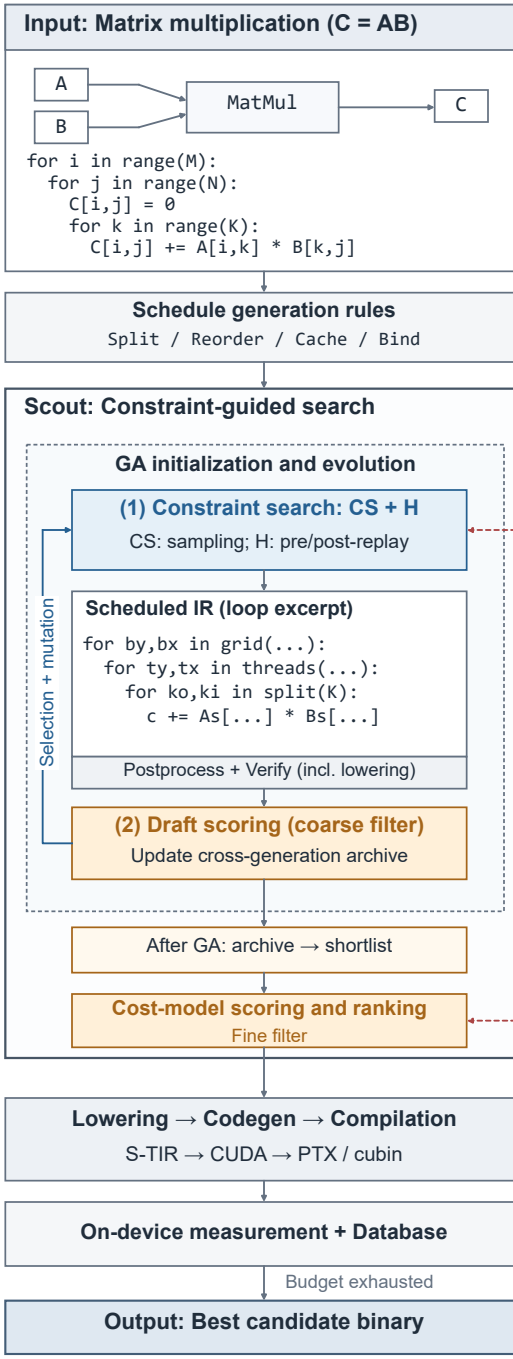

Input: Matrix multiplication (C = AB)
A
B
MatMul
C
for i in range(M):
for j in range(N):
C[i,j] = 0
for k in range(K):
C[i,j] += A[i,k] * B[k,j]
Schedule generation rules
Split / Reorder / Cache / Bind
Scout: Constraint-guided search
GA initialization and evolution
(1) Constraint search: CS + H
CS: sampling; H: pre/post-replay
Scheduled IR (loop excerpt)
for by,bx in grid(...):
for ty,tx in threads(...):
for ko,ki in split(K):
c += As[...] * Bs[...]
Postprocess + Verify (incl. lowering)
(2) Draft scoring (coarse filter)
Update cross-generation archive
Selection + mutation
Measurement feedback
After GA: archive → shortlist
Cost-model scoring and ranking
Fine filter
Lowering → Codegen → Compilation
S-TIR → CUDA → PTX / cubin
On-device measurement + Database
Budget exhausted
Output: Best candidate binary